

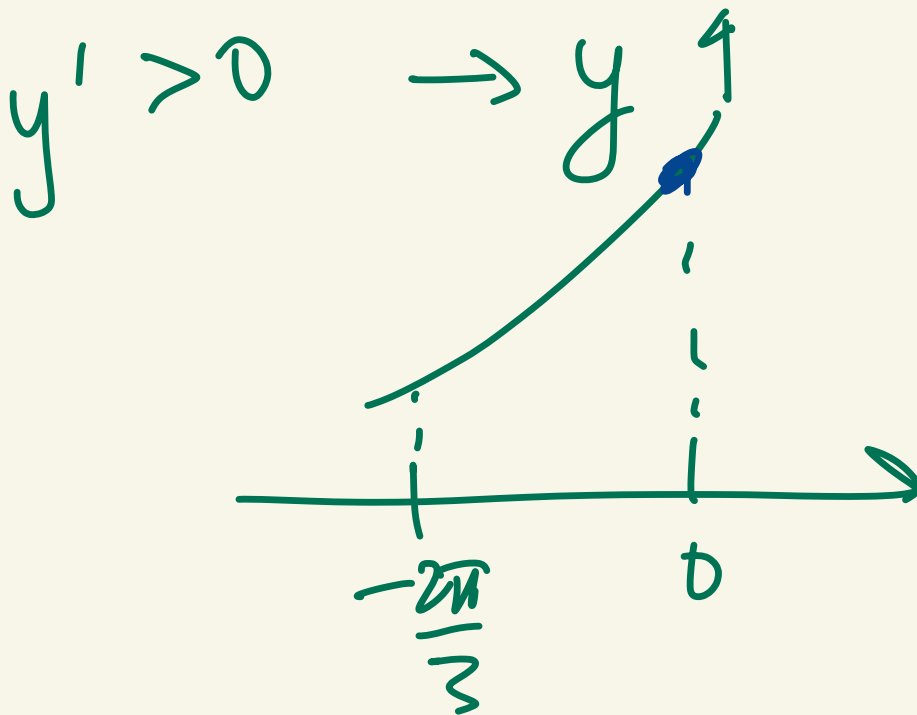


$$y = 6 \cos x + \frac{24}{\pi} x + 5$$

max. str. ?
 $[-\frac{2\pi}{3}; 0]$

$$y' = -6 \sin x + \frac{24}{\pi} \neq 0$$

$$\sin x = \frac{24}{\pi} \cdot \frac{1}{6} = \frac{4}{\pi} \quad \frac{4}{3,14} > 1$$



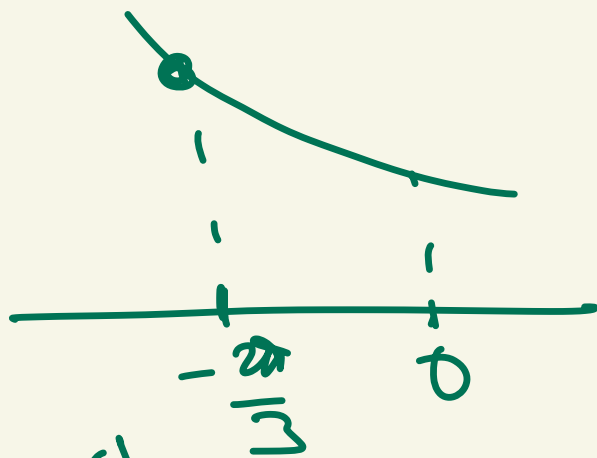
$$y(0) = 6 \cdot \cos 0 + \frac{24}{\pi} \cdot 0 + 5 = \underline{\underline{11}}$$

$$y = 2 \cos x - \frac{18x}{\pi} + 4$$

$\left[-\frac{2\pi}{3}; 0\right]$
наиб. зп

$$y' = \frac{-2 \sin x}{\sin x} - \frac{18}{\pi} < 0$$

$$\sin x = -\frac{9}{\pi}$$

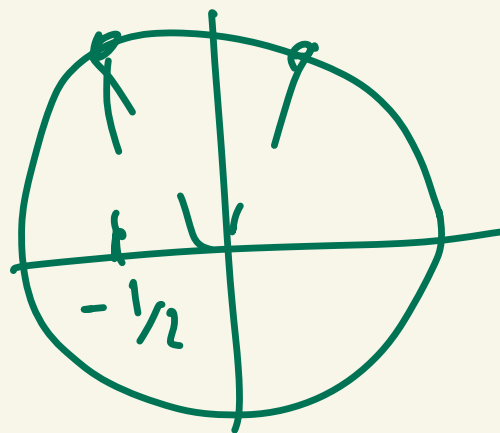


$$y = 2 \cos\left(\frac{2\pi}{3}\right) - \frac{18}{\pi} \left(-\frac{2\pi}{3}\right) + 4$$

$$2 \cdot \left(-\frac{1}{2}\right) -$$

$$-1 + 12 + 4$$

$$= 15$$



$$y = 16 \cos x - \frac{102}{\pi} x + 41$$

$\left[-\frac{2\pi}{3}; 0\right]$
наиб. зп

$$y' = \underbrace{-16 \sin x}_{\leq 1} - \frac{102}{\pi}$$

$$-1 \leq 16 \sin x - \frac{102}{\pi} < 0$$

$$\sin x = -\frac{102}{16\pi}$$

невозможно
корней нет

$$16 \sin x - \frac{102}{\pi}$$

$$y = 108 \ln x - \frac{36}{\pi} x + \pi$$

наиб. 3π

$$\left[-\frac{5\pi}{6}; 0 \right]$$

$$y' = 10 \cos x - \frac{36}{x}$$

$$10 - \frac{36}{x} \approx 10 - 12 = -2$$

$$\sin\left(-\frac{5\pi}{6}\right) = -\frac{1}{2}$$

$$\underbrace{10 \cdot \frac{1}{2}}_5 \ominus \underbrace{\frac{36}{x} \left(-\frac{5\pi}{6}\right)}_{+30} + \frac{7}{7} = 42$$

$$y = \underbrace{3}_6 - \underbrace{\frac{5\pi}{4}}_0 + 5x - 5\sqrt{2} \sin x$$

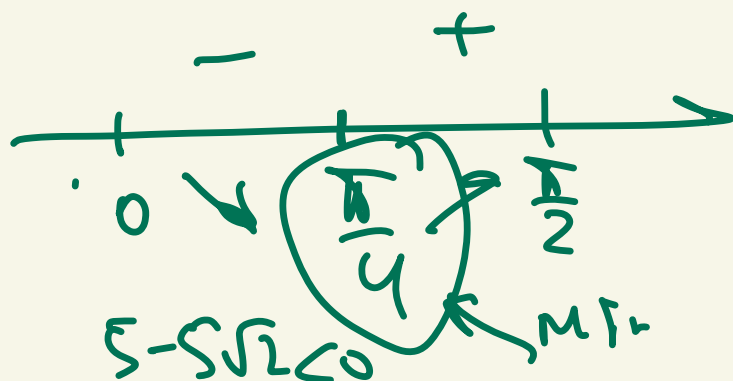
$$5 - 5\sqrt{2} \cos x \left[0; \frac{\pi}{2}\right]$$

Намн. зн?

$$5 - 5\sqrt{2} \cos x = 0$$

$$\cos x = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \Rightarrow x = \frac{\pi}{4}$$

$$\frac{1}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$



$$y = 12 \cos x + 6\sqrt{3}x - 2\sqrt{3}\pi + 6$$

находим π
 $\left[0; \frac{\pi}{2}\right]$

$$y = 3x - 3 \lg x - 5$$

находим π
 $\left[0; \frac{\pi}{4}\right]$

$$\lg'(x) = \frac{1}{\cos^2 x}$$

$$\lg''(x) = -\frac{1}{\sin^2 x}$$

$$\left(\log(x)\right)' = \left(\frac{\sin x}{\cos x}\right)' = \frac{\cos x \cdot \cos x - (\sin x) \cdot \sin x}{\cos^2 x}$$

$$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x}$$

$$\text{ch}'x = \frac{-\sin^2 x - \cos^2 x}{\sin^2 x} = -\frac{(\sin^2 x + \cos^2 x)}{\sin^2 x}$$

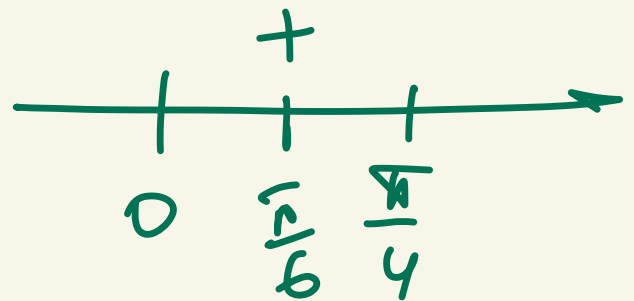
$$= -\frac{1}{\sin^2 x}$$

$$y' = 3 - \frac{3}{\cos^2 x} = \frac{3 \cos^2 x - 3}{\cos^2 x} = 0$$

$$\cos^2 x = 1$$

$$x = 0$$

$$3 - \frac{3 \cdot 2}{2} = 3$$



$$y = \underbrace{(2x-3)\cos x}_{\text{t. max}} - 2\sin x + 2$$

(0, 2\pi)

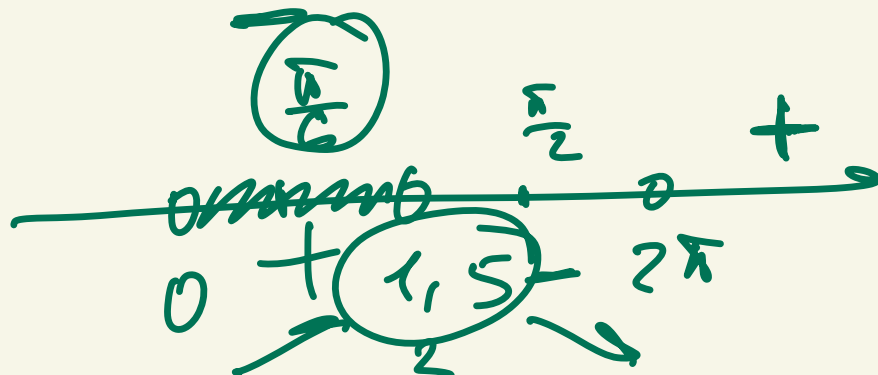
$$y' = 2 \cdot \cancel{\cos x} + (-\sin x)(2x-3) - 2 \cancel{\cos x}$$

$$\underbrace{-\sin x(2x-3)}_{=0} = 0$$

$$\sin x = 0 \quad \text{um} \quad 2x-3=0$$

$$x = 0$$

$$x = 1,5$$



$$\begin{matrix} -\frac{1}{2} \\ \frac{1}{2} \end{matrix} \begin{pmatrix} \frac{2\pi}{6} \\ \frac{\pi}{2} \end{pmatrix} - 3 =$$

$$= -1 \begin{pmatrix} \frac{2\pi}{6} - 3 \\ \frac{\pi}{2} - 3 \end{pmatrix}$$

3,14-3

$$-\sin x \left(\underline{\underline{3-2x}} \right)$$

$\frac{\pi}{6} \qquad \frac{\pi}{2}$

$$-\frac{1}{2} \left(3 - 2 \cdot \frac{\pi}{6} \right)$$

$$\begin{aligned} &\Rightarrow 3 - \frac{\pi}{3} \\ &3 - 1, \dots \\ &\quad \text{+} \end{aligned}$$

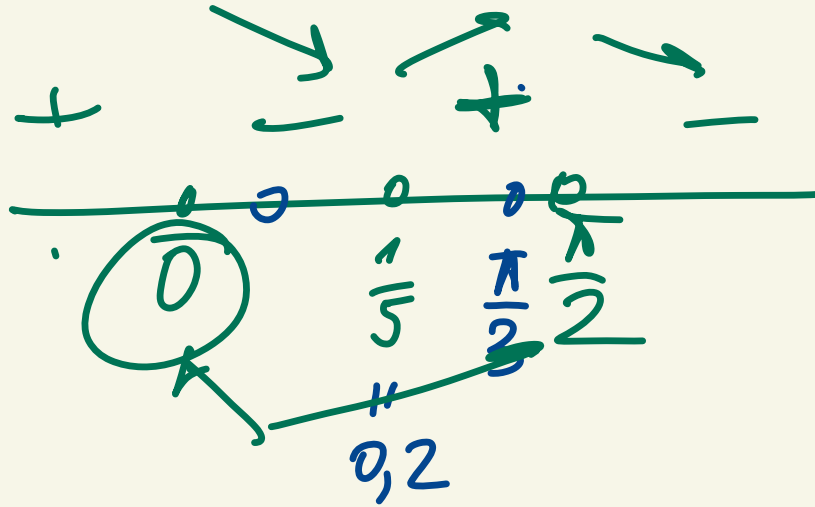
$$y = (1-5x) \underline{\cos x} + 5 \sin x + 3$$

$$T. \max. \left(0 ; \frac{\pi}{2} \right)$$

$$-5 \cos x - \sin x (1-5x) + 5 \cancel{\cos x} + 0$$

$$-\sin x (1-5x) = 0$$

$$x = \frac{1}{5}$$



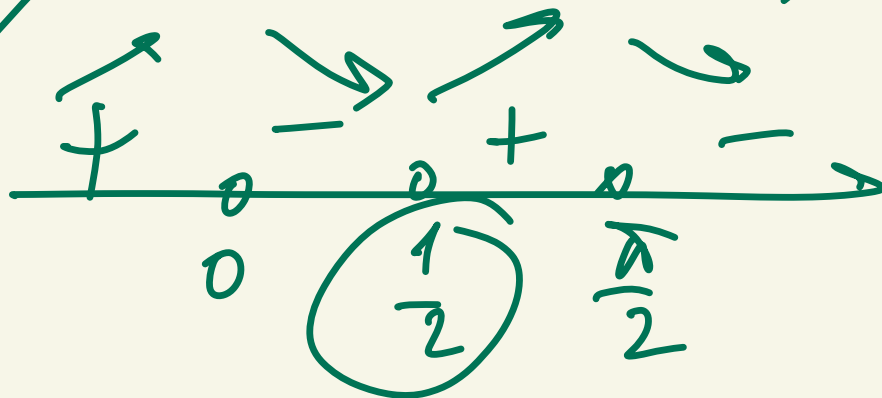
$$-\frac{\sqrt{2}}{2} \left(1 - \frac{5\pi}{3} \right)$$

$$y = (1 - 2x) \cos x + 2 \sin x + 4$$

T. M. M.

$(0; \frac{\pi}{2})$

$$-2 \cos x - \sin x (1 - 2x) + 2 \cos x$$



$$y = 2 \ln x - 4x + \pi - 3$$

$$\left[-\frac{\pi}{3} ; \frac{\pi}{3} \right]$$

$$y' = 2 \frac{1}{\cos^2 x} - 4 = 0$$

$$\frac{2 - 4 \cos^2 x}{\cos^2 x} = 0$$

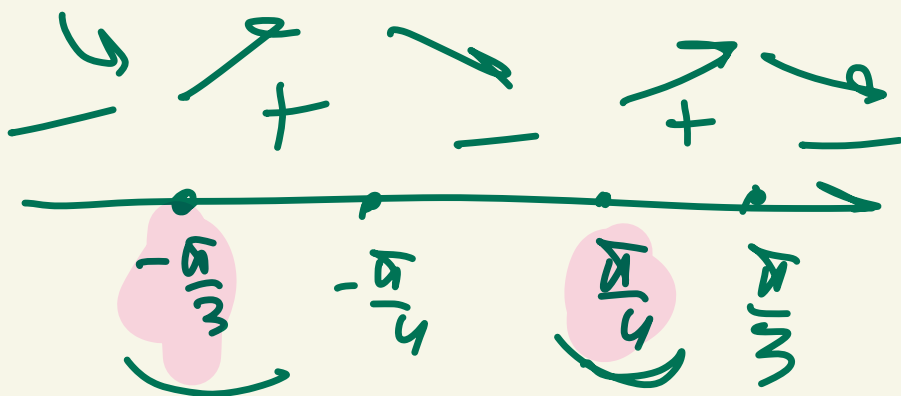
$$\cos^2 x = \frac{1}{2}$$

$$\cos x = \pm \frac{1}{\sqrt{2}}$$

$$\cos x = \pm \frac{\sqrt{2}}{2}$$

$$x = \frac{\pi}{4}$$

$$x = -\frac{\pi}{4}$$



$$y\left(-\frac{\pi}{3}\right) = 2 \lg\left(-\frac{\pi}{3}\right) - \underbrace{4\left(-\frac{\pi}{3}\right)} + \pi - 3$$

$$y\left(\frac{\pi}{4}\right) = 2 \lg\left(\frac{\pi}{4}\right) - 4\left(\frac{\pi}{4}\right) + \pi - 3$$

$$2 - \pi + \pi - 3 = \boxed{-1}$$

$$y = 66 \lg x - 132x + 33\pi + 7$$

$$\text{исслед. на } \left[-\frac{\pi}{3}; \frac{\pi}{3}\right]$$

$$y = \lg_3(x^2 - 6x + 10) + 2$$

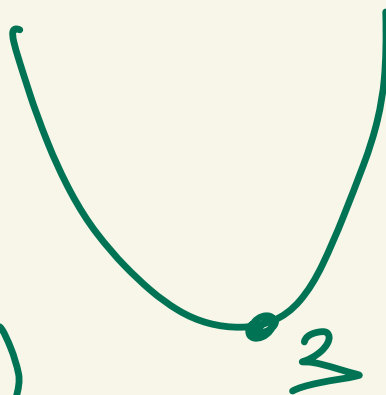
$$y' = \left(\frac{\ln(x^2 - 6x + 10)}{\ln 3} \right)' + 2 \dots$$

$$y' = 2x - 6 = 0 \quad x = 3$$

min

$$y(3) =$$

$$\log_3(9 - 18 + 10)$$



$$y = 7^{x^2 - 2x + 3}$$

$$-\frac{6}{2} = -3$$

log

$$\left(\frac{1}{3}\right)$$

$$(x^2 + 6x + 12)$$

$$9 - 18 + 12$$

$$= 3$$

$$\log_{\frac{1}{3}} 3 = -1$$



$$y = \sqrt{4 - 4x - x^2}$$

т. max

y'

$$-4 - 2x = 0$$

$$\underline{\underline{x = -2}}$$

$$y = \log_5 (\underline{\underline{x^2 - 6x + 12}}) + 2$$

т. min?

$$2x - 6 = 0$$

$$x = 3$$

наши зн

$$y = \sqrt{x^2 - 6x + 13}$$

$$y = 3x - \ln((x+3)^3)$$

$$y' = 3 - \frac{1}{(x+3)} \cdot 3(x+3)^2$$

$$((x+3)^3)' = 3(x+3)^2 \cdot 1$$

Meinung: 3 n.

$$y = \ln(x+4)^9 - 9x$$

$$[-3,5; 0]$$

$$y' = \frac{1}{(x+4)^9} \cdot 9(x+4)^8 \cdot 1 - 9$$

$$y' = \frac{9}{x+4} - 9 = 0$$

~~y~~

$$x = -3$$

$$y = \ln(\underbrace{-3+4})^9 - \underbrace{9(-3)} = 27$$

$$\cos x = \frac{2}{6}$$

$$\frac{3}{2}\sqrt{x} - 18 = 0 \quad \sqrt{x} = \frac{18 \cdot 2}{3}$$

1. Найдите наименьшее значение функции $y = x^{\frac{3}{2}} - 18x + 15$ на отрезке $[3; 410]$.

$$\frac{3}{2}x^{\frac{3}{2}-1} = \frac{(12^2)^{\frac{3}{2}}}{x} - 18 \cdot 144$$

2. Найдите точку минимума функции $y = -\frac{x}{x^2 + 676}$.

3. Найдите наибольшее значение функции $y = 2\sqrt{2}\sin x - 2x + 0,5\pi + 21$ на отрезке $[0; \frac{\pi}{2}]$.

$$2\sqrt{2}\cos x - 2 = 0 \quad \cos x = \frac{1}{\sqrt{2}}$$

4. Найдите точку максимума функции $y = (x^2 - 10x + 10)e^{5-x}$.

5. Найдите наименьшее значение функции $y = 10x - 10\ln(x+3) + 24$ на отрезке $[-2, 5; 0]$.

$$\frac{x^2 - 676}{(x^2 + 676)^2} = 0$$

$$6. y' = 10 - \frac{10}{x+3}$$

Найдите наибольшее значение функции $y = 9^{-34-12x-x^2}$.

$$-2x - 12 = 0$$

7. Найдите наименьшее значение функции $y = 5 + 9x - \frac{x^3}{3}$ на отрезке $[-3; 3]$.

$$9^{-3}$$

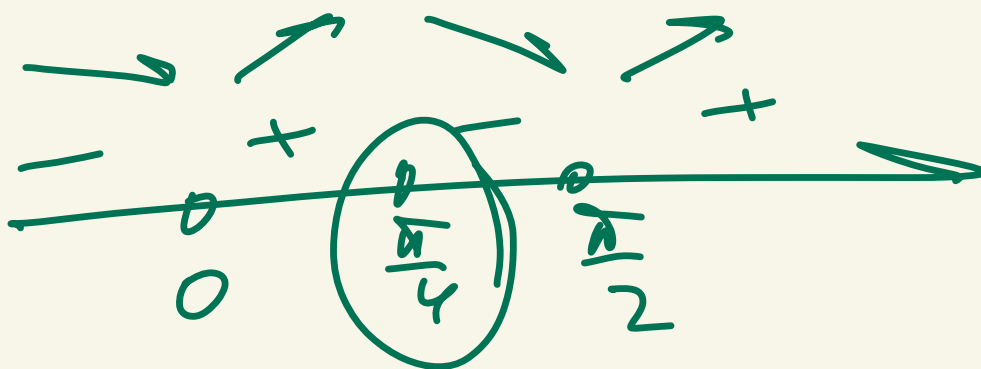
8. Найдите наименьшее значение функции

$y = 3\sin x + \frac{30}{\pi}x + 3$ на отрезке $[-\frac{5\pi}{6}; 0]$.

$$12^3 - 18 \cdot 144 + 15$$

$$1728 - 2592 + 15$$

$$-864 + 15 = \underline{\underline{-849}}$$



$$2\sqrt{2} \cos x - 2$$

$$\sqrt{2} - 2$$

$$y' = (2x - 10) e^{5-x} + e^{5-x} (-1) \cdot (x^2 - 10x + 10)$$

$$e^{5-x} (2x - 10 - x^2 + 10x - 10)$$

$$-x^2 + 12x - 20$$

